ASA Toxicity

Section I: Scenario Demographics

Scenario Title:	ASA overdose
Date of Development:	05/02/2015 (DD/MM/YYYY)
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a febrile, altered patient resulting from ASA toxicity.
CRM Objectives:	1) Demonstrates situational awareness by avoiding fixation error and re-assessing and re-evaluating a changing clinical picture 2) Adequately utilizes resources through early call for help and appropriate delegation of tasks 3) Performs a tailored and appropriately brief goals of care discussion with the family of a critically unwell patient
Medical Objectives:	1) Initiates broad work-up of the febrile altered patient 2) Identifies AG metabolic acidosis on critical VBG/lytes 3) Administers appropriate early treatment for ASA toxicity 4) Employs a tailored approach to intubation and ventilator settings 5) Recognizes the need for dialysis

Case Summary: Brief Summary of Case Progression and Major Events
The learner will be presented with an altered febrile patient, requiring an initial broad work-up and management plan. The learner will receive a critical VBG report of severe acidosis, hypoglycemia and hypokalemia, requiring management. Following this, the rest of the blood work and investigations will come back, giving the diagnosis of salicylate overdose. The patient's mental status will continue to decline and learners should proceed to intubate the patient, anticipating issues given the acid-base status. The learner should also initiate urinary alkalinization and make arrangements for urgent dialysis.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.

Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient	*Select most important dimension(s)*	☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates	Brief Description of Role
Daughter	Provides collateral history. Last seen well 3 days ago. Full code. After results are back, confirms recent stressor being left at alter for 2nd marriage
Nurse	To assist at bedside, cue learners to patient's respiratory status and seizure PRN

B. Required Monitors		
☒ EKG Leads/Wires	☒ Temperature Probe	☒ Central Venous Line
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

C. Required Equipment		
☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☒ Other: Dialysis catheter kit
☐ Intraosseous Set-up	☐ LMA	☐ Other:

D. Moulage
Old man mask.

E. Approximate Timing					
Set-Up:	20 min	Scenario:	15 min	Debriefing:	30 min

Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case
You are working at a community hospital. The triage nurse comes to tell you that they have just put an 82 year-old male in a resuscitation room. He was found unresponsive by daughter and was brought in by EMS. In triage he was profoundly altered, febrile and hypotensive. His daughter is in the room with him.

B. Patient Profile and History				
Patient Name: John Valentine		Age: 82	Weight: 75 kg	
Gender: ☒ M ☐ F		Code Status: Full		
Chief Complaint: Altered LOC				
History of Presenting Illness: Last seen well 3 days ago. He was not answering the phone so daughter went to his house today and found him unresponsive in his bed. Previously well. Was supposed to get re-married about a week ago, but she left him at the altar. He's been pretty down since then, but nothing unexpected.				
Past Medical History:	HTN	Medications:	Metoprolol 50mg BID	
	Dyslipidemia		Crestor 40mg OD	
	Hypothyroidism		Levothyroxine 100mcg OD	
	Depression		Citalopram 20mg OD	
Allergies: None				
Social History: Lives alone. Non-smoker. Non-drinker				
Family History: Nil				
Review of Systems:	CNS:	Daughter unaware of previous complaints		
	HEENT:	"		
	CVS:	"		
	RESP:	"		
	GI:	"		
	GU:	"		
	MSK:	"	INT:	

C. Baseline Simulator State and Physical Exam			
☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display	
HR: 122/min	BP: 83/60	RR: 35/min	O_2SAT: 92 % NRB
Rhythm: NSR	T: 38.6°C	Glucose: 3.1mmol/L	GCS: 9 (E3 V2 M4)
General Status: Unresponsive on bed			

CNS:	Opens eyes to voice. Confused. Withdraws to pain bilaterally.		
HEENT:	No signs HI. No meningismus. Pupils 2 mm fixed bilaterally.		
CVS:	No murmur. No edema.		
RESP:	Agonal respirations. Notable tachypnea and work of breathing, bilateral crackles.		
ABDO:	Soft. Normal bowel sounds.		
GU:	Nil		
MSK:	No signs trauma.	SKIN:	Dry

Section VI: Scenario Progression

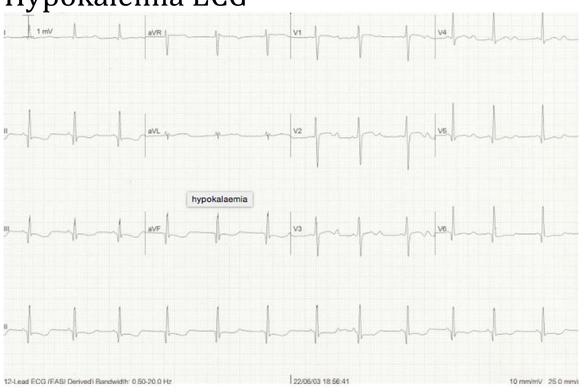
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 122/min BP: 83/60 RR: 35/min O_2SAT: 93% NRB T: 38.6°C	GSC 9 (E3 V2 M4) **When asked re: goals of care, daughter indicates "would want everything done"** **RN to cue re: high resp rate, increased WOB**	<u>Learner Actions</u> - ☐ IV, O2, monitors - ☐ Fluid bolus - ☐ Check glucose **(3.1)** and replace with D50 - ☐ Tox + Cardiac + Sepsis + CK + TSH blood work - ☐ ECG - ☐ Portable CXR - ☐ Broad spectrum Abx - ☐ Obtain collateral history and goals of care from daughter - ☐ CT Head when stable	<u>Modifiers</u> *Changes to patient condition based on learner action* - D50 given → no effect on GCS - 1st fluid bolus → BP 95/70, HR 108 - 2nd bolus → BP 100/78, HR 97 <u>Triggers</u> *For progression to next state* - Capillary glucose not checked by 3 minutes → **3. Seizure** - Fluids given, labs ordered, glucose corrected, or 5 min. → **2. VBG Back**
2.VBG Back Rhythm: NSR HR: 100/min BP: 97/72 RR: 40/min O_2SAT: 90 % NRB	**Give VBG ONLY at onset of state.**	<u>Learner Actions</u> - ☐ Treat hypokalemia - ☐ Recheck glucose **(3.2)** - ☐ Start D5 infusion - ☐ Recognize AG metabolic acidosis	<u>Triggers</u> - Capillary glucose not rechecked by 3 mins → **3. Seizure** - Lytes, glucose corrected → **4. Rest of Labs Back**
3. Seizure Rhythm: NSR HR: 150/min BP: 100/75 RR: 18/min O_2SAT: 85 %	Patient begins to seize **(RN to cue)**	<u>Learner Actions</u> - ☐ Check glucose **(1.9)** - ☐ D50 bolus - ☐ IV benzodiazepine	<u>Modifiers</u> <u>Triggers</u> - From state 1. & glucose corrected → **2. VBG back** - From state 2. & glucose corrected → **4. Rest of Labs back**
4. Rest of Labs Back Rhythm: NSR HR: 95 BP: 95/65 RR: 42 O2: 85% NRB	GCS 5, patient vomiting. (**RN to prompt that patient not protecting, more somnolent**) **Give remaining lab results at onset of state.**	<u>Learner Actions</u> - ☐ Update daughter - ☐ Initiate urinary alkalization - ☐ Nephrology & ICU Consult - ☐ Intubation with plan re: acidosis (NaHCO3 before) - ☐ Post-intubation care (set vent to high RR, sedation, CXR) - ☐ Call Poison Control	<u>Modifiers</u> <u>Triggers</u> - Considers acidosis with intubation plan → **END CASE PRN** - Does not consider acidosis in intubation plan (does not match vent to RR) → **5. VF Arrest**
5. VF Arrest	Patient loses pulse with VF rhythm x 3 rounds	<u>Learner Actions</u> - ☐ Good quality CPR - ☐ Shock VF - ☐ Epinephrine - ☐ Amiodarone	<u>Triggers</u> - 15 min → **End Case PRN**

Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 135	K: 2.0	Cl: 100	HCO_3: 10	BUN: 20	Cr: 580	Glu: 3.3
Ca: 2.58	Mg: 0.65 (low)	PO_4: 0.80	Albumin: 12			
AG = 27	OG = 7	CK: 500	TSH: 0.4 (normal)			
VBG	pH: 7.05	PCO_2: 30	PO_2: 90	HCO_3: 8	Lactate: 10	
Troponin: 7	INR: 3.3	PTT: 40				
WBC: 18	Hg: 120	Hct: 0.52	Plt: 330			
AST: 300	Tylenol: <0.15	EtOH: <2	ASA: 7.6	Osm: 300		

Images (ECGs, CXRs, etc.)

Hypokalemia ECG

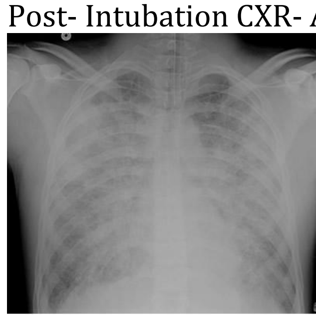


ECG source: https://lifeinthefastlane.com/ecg-library/basics/hypokalaemia/

Pre-Intubation CXR- ARDS

CXR source: http://www.radiology.vcu.edu/programs/residents/quiz/pulm_cotw/PulmonConf/09-03-04/68yM%2008-03-04%20CXR.jpg

Post- Intubation CXR- ARDS



CXR source: http://courses.washington.edu/med620/images/mv_c3fig1.jpg

Ultrasound Video Files (if applicable)	
Abdomen: no FF	Aorta: no AAA
Cardiac: no PCE	

Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video

Objectives	
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Sample Questions for Debriefing

1. What is your differential diagnosis for the altered and febrile patient?
2. What is on the differential diagnosis for an AG metabolic acidosis? For an Osmolar Gap acidosis? For a double gap acidosis?
3. What acid-base disturbances are seen with ASA overdose? Why?
4. What are the metabolic effects of an ASA overdose?
5. What is the toxic dose of ASA? When do we do levels?
6. List 3 causes of SOB/tachypnea in the ASA toxic patient
7. What is the goal urine pH and serum pH during urinary alkalinization?
8. What are the indications for urinary alkalinization?
9. What are the indications for dialysis?

Key Moments
Initial code discussion and update with reconsideration of code status given OD diagnosis
Initiate urinary alkalinization
Recognize need for dialysis